

08.01 Modeling with Exponential Equations and Expressions

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
Exponential And Logarithmic Models	Decide whether the unknown is in an exponent, inside a logarithm, or in a model parameter.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

A. Practice: Modeling with Exponential Equations and Expressions

1.

$$A(t) = 500(1.06)^t$$

2.

$$3^x = 20$$

3.

$$\log_b \left(\frac{x^3}{y} \right)$$

4.

$$A(t) = 500(1.06)^t$$

5.

$$3^x = 20$$

6.

$$\log_b \left(\frac{x^3}{y} \right)$$

7.

$$A(t) = 500(1.06)^t$$

8.

$$3^x = 20$$

9.

$$\log_b \left(\frac{x^3}{y} \right)$$

10.

$$A(t) = 500(1.06)^t$$

11.

$$3^x = 20$$

12.

$$\log_b \left(\frac{x^3}{y} \right)$$

13.

$$A(t) = 500(1.06)^t$$

14.

$$3^x = 20$$

15.

$$\log_b \left(\frac{x^3}{y} \right)$$

16.

$$A(t) = 500(1.06)^t$$

17.

$$3^x = 20$$

18.

$$\log_b \left(\frac{x^3}{y} \right)$$

19.

$$A(t) = 500(1.06)^t$$

20.

$$3^x = 20$$

B. Apply the key formulas

1.

$$A(t) = A_0 b^t$$

2.

$$\log_b (b) = y \iff b^y = b$$

3.

$$\log_b (MN) = \log_b M + \log_b N$$

4.

$$A(t) = A_0 b^t$$

5.

$$\log_b (e) = y \iff b^y = e$$

6.

$$\log_b (MN) = \log_b M + \log_b N$$

7.

$$A(t) = A_0 b^t$$

8.

$$\log_b (c) = y \iff b^y = c$$

9.

$$\log_b (MN) = \log_b M + \log_b N$$

10.

$$A(t) = A_0 b^t$$

11.

$$\log_b (a) = y \iff b^y = a$$

12.

$$\log_b (MN) = \log_b M + \log_b N$$

13.

$$A(t) = A_0 b^t$$

14.

$$\log_b (d) = y \iff b^y = d$$

15.

$$\log_b (MN) = \log_b M + \log_b N$$

16.

$$A(t) = A_0 b^t$$

17.

$$\log_b (b) = y \iff b^y = b$$

18.

$$\log_b (MN) = \log_b M + \log_b N$$

19.

$$A(t) = A_0 b^t$$

20.

$$\log_b (e) = y \iff b^y = e$$